GG-T-356a
September 25, 1967
SUPERSEDING
Fed. Spec. GG-T-356
September 19, 1957

FEDERAL SPECIFICATION

THERMOMETER, SELF-INDICATING, LIQUID IN GLASS, MINIMUM, (METEOROLOGICAL)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 This specification covers the requirements for one type of alcohol-in-glass thermometer, minimum registering.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein

Federal Specifications:

QQ-S-766 - Steel Plates, Sheets, and Strip —Corrosion Resisting.
PPP-B-636 - Box, Fiberboard.
PPP-B-665 - Boxes, Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676 - Boxes, Setup.
PPP-C-843 - Cushioning Material, Cellulosic.
PPP-F-320 - Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-T-76 - Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing).
PPP-T-97 - Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment, (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-P-116 - Preservation, Methods of.

Military Standards

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6660

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:50Z
Check the source to verify that this is the current version before use.

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards (NBS) Handbook:

H-28 Screw Thread Standards for Federal Services.

(Copies may be obtained upon application accompanied by money order, coupon, or cash to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C.,20402. Price may be obtained for the Superintendent of Documents.)

3. REQUIREMENTS

3.1 Parts, materials, and processes. Parts, materials, and processes not described in this specification shall conform to the requirements specified on figures 1 (back), 2 (tube), 3 (clamp), and 4 (assembly). Unspecified parts, materials, and processes shall be selected by the contractor so that they will be entirely suitable for the intended purpose.

3.2 Construction. Thermometers shall be constructed in accordance with this specification, including the figures specified in 3.1. All construction shall be such as to meet the inspection and test requirements of section 4, as applicable, and perform satisfactorily under normal use when mounted with the bore forming an angle of $5^{\circ} \pm 2^{\circ}$ with the horizontal and the bulb at the lower end.

3.3 Glass. All glass shall be thoroughly annealed or seasoned in accordance with the best practice of the manufacturers of meteorological thermometers.

3.4 Alcohol. All alcohol shall be chemically pure grain alcohol. Coloring agent (5 gm. of crystal iodine to 1000 cc. of 190 proof alcohol) shall remain in solution at all temperatures from -90° to 145° F. At the discretion of the manufacturer the space above the alcohol shall be filled with an inert gas. The bore and bulb shall be free of foreign matter such as chips of glass, dust, or dirt. The method of filling and pressure employed shall conform to the best thermometer filling practice.

3.5 Tube. The tube shall consist of a cylindrical glass stem with a round or oval capillary bore, and a glass bulb sealed on the end of the stem. Any evidence of strain, either ring or longitudinal, shall be faint and highly diffused.

3.5.1 Stem. The stem shall be made of good quality thermometric glass of a type suitable for the range of temperature to be measured. The stem shall have a strip of opalescent glass incorporated therein, running the full length of the stem and subtending not less than 90° of the circumference. This strip shall be positioned with respect to the scale graduations to provide a background to facilitate reading. All stems shall be etched to provide the markings specified in 3.5.1.1 and 3.5.1.2. Intervals between scale graduations shall be 1° on either Fahrenheit or centigrade scales. All lines, figures, and letters shall be clear-cut and distinct, and filled with the best quality black pigment which will not wash out in water or alcohol when tested in accordance with 4.4.1.1.3. All etching shall be visible without removing the tube from the back.

3.5.1.1 Stem markings. Each tube shall be marked near the upper end and to the right of the scale with a serial number to be designated when the order is placed. A symbol of the procuring agency shall precede the serial number. (Example: U. S. Weather Bureau will be USWB-Serial No.---, as per detail A, fig. 2.)

3.5.1.2 Scale designations. Suitable identification for the appropriate scale, either "F" or "C," shall be etched near the top of the stem as shown in detail A, fig. 2.

3.5.1.3 Nib. A terminal nib drawn from the end of the stem and not merely fused to the stem, shall be provided as shown in detail B of figure 2 to maintain alignment of the stem with, and positioning in, the corrugation in the back on which it is mounted. The terminal nib shall be smoothly formed and shall fit snugly into the hole provided for it in the back, in such a way that the thermometer lies flush with the back. The nib shall be so positioned relative to the scale graduations that these graduations will be faced directly outward from the back.

3.5.2 Bulb. The bulb shall be made of Corning normal glass, or equal. It shall be spherical in shape and smoothly and neatly joined to the lower end of the stem and shall act as a reservoir for the liquid. It shall be sensibly concentric with the stem. The wall of the bulb shall be blown as thin as possible consistent with necessary strength, in order to reduce the time lag of the thermometer to a minimum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:50Z
Check the source to verify that this is the current version before use.

3.5.3 Expansion chamber. The thermometer shall be protected against breakage, when exposed to temperatures above the normal range, by means of an expansion chamber. This protection shall be effective to either 135° or 160° F., as specified by the procuring agency (see 6.2). The design of the expansion chamber shall be left to the discretion of the manufacturer. The design shall be such as to meet the test for resetting specified in 4.4.2. The range scale length shall be not less than 7-1/2 inches.

3.5.4 Index. The index used to provide for self-registration shall be made of black glass and shall be loosely fitted into the bore of the stem. The index shall be slightly longer than the diameter of the bulb, and shall be straight, smoothly formed, and provided with rounded terminal knobs. It shall retreat with the alcohol surface when the temperature falls, but shall permit the alcohol to flow past it when the temperature rises.

3.6 Range requirements. Unless otherwise specified, the range requirements for the thermometer shall conform to table I. The range shall be as specified by the procuring agency (see 6.2).

TABLE I. Range requirements

Range (°F.)	Range (°C)
-5 to + 135	-20 to + 60
-20 to + 120	-30 to + 50
-40 to + 110	-40 to + 45
-90 to + 90	-70 to + 35

3.6.1 Special ranges. When need arises, other ranges may be ordered not to exceed a span of 180°F or 100° C.

3.7 Accuracy. The accuracy of the thermometer shall conform to table II. (See 4.4.1.2.)

TABLE II. Accuracy

Range (°F.)	Tolerance (°F.)	Range (°C.)	Tolerance (°C.)
0 and above	± 0.4	-18 and above	± 0.2
-50 to 0	± 1.0	-45 to -18	± 0.5
Below -50	± 1.5	Below -45	± 0.8

3.7.1 Accumulative scale error The change in error over a span of 20° F or 10° C shall not exceed the tolerances specified in 3.7.

3.7 2 Check points. At least five temperature check points shall be used by the contractor in the manufacture of the thermometer. These five check points shall be separated by at least 20° F or 10° C.

3.8 Interchangeability. All parts of the same type shall be physically and mechanically interchangeable.

3.9 Cleaning. After metal parts have been fabricated they shall be cleaned in accordance with the best commercial practice.

3.10 Preproduction samples.

3.10.1 Quantities. When required by the procuring agency, five preproduction samples shall be furnished for Government inspection and approval (see 6.2)

3.10.2 Compliance with specific requirements.

3.10.2.1 Preproduction samples shall meet specified requirements, shall be made and assembled by tools and methods that will be used for quantity production, and shall be accompanied by a statement to that effect except as specified in 3.10.2.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:50Z
Check the source to verify that this is the current version before use.

3.10.2.2 Approval of preproduction samples shall not be construed as a waiver of any specified requirement. After being released to the contractor (see 3.10.4), preproduction samples to be offered as units on contract shall be refabricated by the contractor, if necessary, to meet specified requirements.

3.10.2.3 When deviation from the requirements of 3.10.2.1 is unavoidable, the preproduction samples may be submitted for approval provided that the accompanying statement describes in detail each nonconforming feature, reason therefor, and manner in which it will be corrected in production of equipment on contract.

3.10.3 Test data. The preproduction samples shall be accompanied by test data showing compliance with specified performance. The test data shall comprise an engineering report giving test procedure, observations and other data, calculations, test results, and essential details of the testing equipment (manufacturer's model, serial number, date of calibration, and the like)

3.10.4 Reference standards. After preproduction samples have been approved and returned to the contractor's plant, they shall be kept intact in custody of the Government inspector until released by him. They shall be used as reference standards to resolve any differences of opinion regarding interpretation of specified requirements.

3.11 Inspection standards. Concurrently with or prior to the submission of the preproduction samples required in 3.10.1, the contractor shall submit a certificate of accuracy from some recognized standardizing agency covering the test standards which he proposes to use in the graduation and inspection of instruments on order. The statement shall include a complete description of the standards, and the most recent test data taken to determine their accuracy Full details of dates, methods used, and conditions under which the data was taken, shall be given. Written approval shall be obtained before using these standards for calibrating or testing the instruments furnished on order. The following types of standards certified at the times indicated, are acceptable:

(a) Mueller Bridge and Platinum Resistance Thermometer· Check once a year for first 3 years; once every 3 years thereafter.

(b) Alcohol-in-glass: The smallest graduations of this thermometer shall be not greater than 0.1° F. Check once a year.

3.12 Workmanship. Components, subassemblies, and parts shall be clean and manufactured and assembled in a thoroughly workmanlike manner.

3.13 Idenfication marking. The thermometer back shall be permanently stamped or engraved in accordance with the marking shown in figure 4.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specificati on where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements

4.2 Preproduction and acceptance inspection (end item).

4.2.1 Preproduction inspection. Preproduction inspection shall consist of group A inspection (see 4.2.2.1), and preproduction inspection as specified in table III. Nondestructive inspection on preproduction samples may be performed to determine compliance with specified requirements.

4.2.2 Acceptance inspection. Acceptance inspection may be performed at the manufacturer's plant or Government laboratory designated by the procuring agency (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:50Z
Check the source to verify that this is the current version before use.

TABLE III. Preproduction inspection

Inspection	Requirement par.	Inspection/test reference	Pre-production inspection/test	AQL (end item) 1/	
				Major	Minor
Visual and mechanical	3.2	4.3	X	1 percent for the group	4 percent for the group
Accuracy of indication	3.7	4.4.1	X		
Expansion chamber	3.5.3	4.4.2 2/	X		

1/ See 4.3.2.
2/ Test for resetting.

4.2.2.1 Group A inspection. Group A inspection (including sampling) shall conform to table III and MIL-STD-105. One hundred percent inspection may be performed at the discretion of the procuring agency. Group A inspection shall be performed in any order which is satisfactory to the Government inspector.

4.2.2.2 Disposition of nonconforming product. When defective sample units on rejected lots are resubmitted for acceptance, such product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault The record shall be presented to the Government when the product is resubmitted and shall become the property of the Government.

4.3 Visual and mechanical inspection (end item and preparation for delivery).

4.3.1 End item.

4.3.1.1 Backs. A visual inspection shall be made of the backs furnished on the order Any back which has imperfect stamping shall be rejected. In addition to the visual inspection, the backs shall be measured for conformance with applicable figures. Backs shall not be formed or bent in any manner which will transfer any undue strain to the tubes when the components are properly assembled.

4.3.1.2 Tubes. The tubes shall be measured to determine that they are within the dimensional tolerances specified in the applicable figures. When specified, the contractor shall supply "Go/No-Go" gages of simple construction to be used in these measurements. (See 6.2.)

4.3.2 Preparation for delivery. Inspection shall be made to determine compliance with the packaging, packing, and marking requirements of section 5. Defects shall be scored in accordance with table IV. For examination of interior packaging, the sample unit shall be one shipping container prepared for delivery and selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers containing thermometers offered for acceptance at one time. The inspection level shall be S-2 with an acceptable quality level (AQL) of 4.0 defects per hundred units.

TABLE IV. Inspection of preparation for delivery

Examine	Defects
Markings (exterior and interior).	Omitted, incorrect, illegible improper size, location, sequence, or method of application.
Materials.	Any component missing or damaged.
Workmanship.	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.
Containers.	Not as specified
Contents.	Quantity in intermediate package and level B shipping container not as specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:50Z
Check the source to verify that this is the current version before use.

4.4 Tests.

4.4.1 Test for accuracy of indication.

4.4.1.1 Testing conditions. In performing the tests of 4.4.1.2, the instruments shall be treated under the following conditions:

4.4.1.1.1 The instruments to be tested shall be immersed in a liquid bath which shall be constantly agitated in order that all instruments may be exposed to the same temperature.

4.4.1.1.2 The appropriate standard thermometer, for which test data has been submitted and approved in accordance with 3.11, shall be immersed in the same bath as the instruments being checked Both the standard thermometer and the test instruments shall have their bulbs placed at approximately the same level in the bath. The instrument shall be totally immersed in the bath

4.4.1.1.3 The temperature of the liquid bath shall be held constant to within 0.1° F. or 0 05° C Data shall not be taken until both the standard thermometer and test instruments have completely responded to the temperature of the bath.

4.4.1.1.4 When requested by the procuring agency, the contractor shall furnish the inspector a statement showing the check points which were used in graduating the thermometer tube. In conducting the test of 4.4.1.2, all test temperatures shall be selected at points intermediate between these check points. (See 6.2.)

4.4.1.1.5 When test data are taken, the line of sight from the observer's eye shall be normal to the stem of the instruments.

4.4.1.2 Test procedure. The instruments shall be compared with the indications of the standard thermometer at not less than four points approximately equally distributed throughout the range of the scale. When the indications of the standard thermometer have been corrected to true temperatures, the differences between these true temperatures and the indications of the instrument being tested shall be within the applicable tolerances given in 3.7. The change in error over a span of 20° F. or 10° C. shall not exceed 0.5° F. or 0.3° C. on any part of the appropriate scale.

4.4.1.2.1 Accumulative scale error. The thermometer shall be tested throughout any 20° F. or 10° C. range between the check points to determine compliance with 3.7.1.

4.4.2 Test for resetting. This test shall be applied to all thermometers which have been tested in accordance with 4.3 and 4.4.1. The thermometers shall be warmed until the columns read approximately 100° F. or 40° C. and the index set at top of column. The thermometers shall then be placed in a horizontal position and the temperature lowered, at a rate not to exceed 25° F. per minute, to approximately the lowest scale reading. If the index sticks at any point along the bore or fails to remain at the lowest temperature reached (the thermometers undisturbed), it shall be considered defective and subject to rejection.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Each thermometer shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Each thermometer shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Unit packaging. Cushion each thermometer by wrapping in cellulosic material conforming to PPP-C-843, type II, class A. Place the cushioned item within a close-fitting fiberboard sleeve. The fiberboard sleeve shall be constructed of material conforming to PPP-F-320, type CF, class weather-resistant, variety SW, grade W6c. The openings on both ends of the sleeve shall be closed with tape conforming to PPP-T-76.

5.1.1.4 Intermediate packaging. Ten thermometers, unit packaged as specified in 5.1.1.3, shall be placed in a fiberboard box conforming to PPP-B-636, W6c. Closure of the box shall be in accordance with the appendix to the box specification. In addition, all seams and joints shall be sealed with tape, not less than 2 inches wide, conforming to PPP-T-76.

5.1.2 Level B.

5.1.2.1 Unit packaging. Each thermometer, cleaned in accordance with the manufacturer's normal

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:50Z
Check the source to verify that this is the current version before use.

practice, shall be packaged as specified in 5.1.1.3, except that the fiberboard shall be class domestic, grade 175, or each thermometer shall be packaged in a paperboard box conforming to PPP-B-665 or PPP-B-676.

5.1.2.2 Intermediate packaging. Ten unit packages shall be intermediate packaged in a fiberboard box conforming to PPP-B-636, class domestic.

5.1.3 Level C. Each thermometer shall be packaged to afford adequate protection against corrosion, deterioration, and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Unless otherwise specified (see 6.2), a quantity of thermometers packaged as specified in 5.1, shall be packed in a close-fitting fiberboard box conforming to PPP-B-636, type CF, class weather-resistant, special requirements use. Closure shall be in accordance with the appendix to the box specification. The shipping containers shall be reinforced with pressure-sensitive filament tape conforming to PPP-T-97, type IV, as specified in the appendix to the box specification.

5.2.2 Level B. Four intermediate packages (40 thermometers) shall be packed in a close-fitting fiberboard box conforming to PPP-B-636, class domestic.

5.2.3 Level C. A quantity of thermometers packaged as specified in 5.1, shall be packed in a shipping container which will afford adequate protection against damage to the container and its contents during shipment from the supply source to the first receiving activity. Shipping containers shall comply with the rules and regulations of common carriers as applicable to the mode of transportation.

5.3 Marking. (See 6.2.)

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The thermometer covered herein is used for the measurement of minimum temperature for meteorological purposes.

6.2 Ordering data. Purchasers should select the preferred options permitted herein (see fig 3, 4.1, and 5.2.1) and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Symbol of procuring agency; and serial numbers to be employed, if required (see 3.5.1.1).
(c) Degree of temperature protection (see 3.5.3 and 4.4.2).
(d) Range required (see 3.6).
(e) Whether preproduction samples are required (see 3.10.1).
(f) Place of acceptance inspection (see 4.2.2).
(g) Where "Go/No-Go" gages are required (see 4.3.1.2).
(h) Whether temperature check points are needed (see 4.4.1.1.4).
(i) Level of packaging and packing and any special marking for shipment required (see 5.1, 5.2, and 5.3).
(j) Whether rough handling test is required.

6.3 Size data. Size limitations cited herein are based on the dimensions of the equipment for which these requirements were established. Requests to exceed these limitations must be supported by data indicating why limitations cannot be met and recommend a revised limitation.

6.4 This specification includes the requirements of MIL-T-11399.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:50Z
Check the source to verify that this is the current version before use.

GG-T-356a

MILITARY INTEREST:

Custodians:

Army - EL
Navy - OS
Air Force - 11

Review activities:

Army - ME
Navy - AS
Air Force - 01

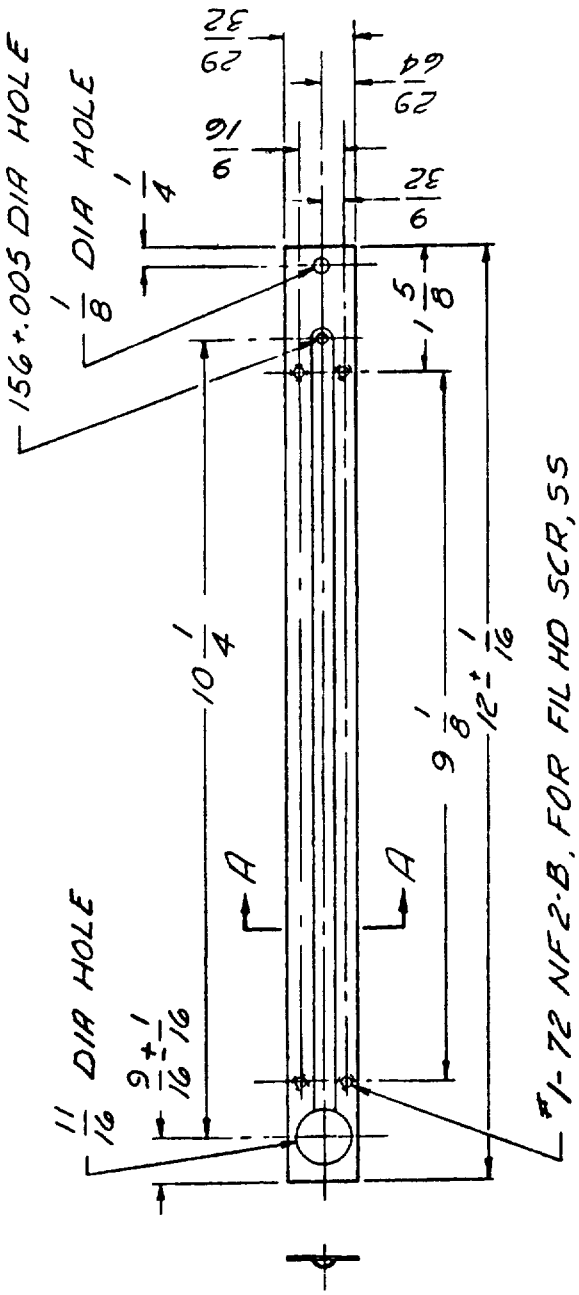
Preparing activity:

Army - EL

CIVIL AGENCIES INTEREST:

U. S. Department of Commerce
GSA-FSS

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:50Z
Check the source to verify that this is the current version before use.



Steel, corrosion resisting, class 4, condition C, temper 2, finish 6, per QQ-S-766. (At the option of the ordering office, when so specified in the order, steel shall be corrosion resisting, class 4, condition a, annealed, finish 6, per QQ-S-766.)

DIMENSIONS IN INCHES UNLESS OTHERWISE SPECIFIED
TOLERANCES ±.005 DECIMALS, ± 1/64 ON FRACTIONS.

FIGURE 1. Back.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:50Z
Check the source to verify that this is the current version before use.

GRADUATION LINES & FIGURES SHALL BE ETCHED ON STEM. GRADUATION LINES SHALL BE STRAIGHT & PERPENDICULAR TO THE AXIS OF THE STEM. GRADUATIONS SHALL BE NUMBERED AT EACH MULTIPLE OF 10 DEGREES, NUMBERS BELOW ZERO TO BE PRECEDED BY A MINUS SIGN. ALL LINES & FIGURES SHALL BE CLEAR CUT & DISTINCT AND SHALL BE FILLED WITH A PERMANENT BLACK.

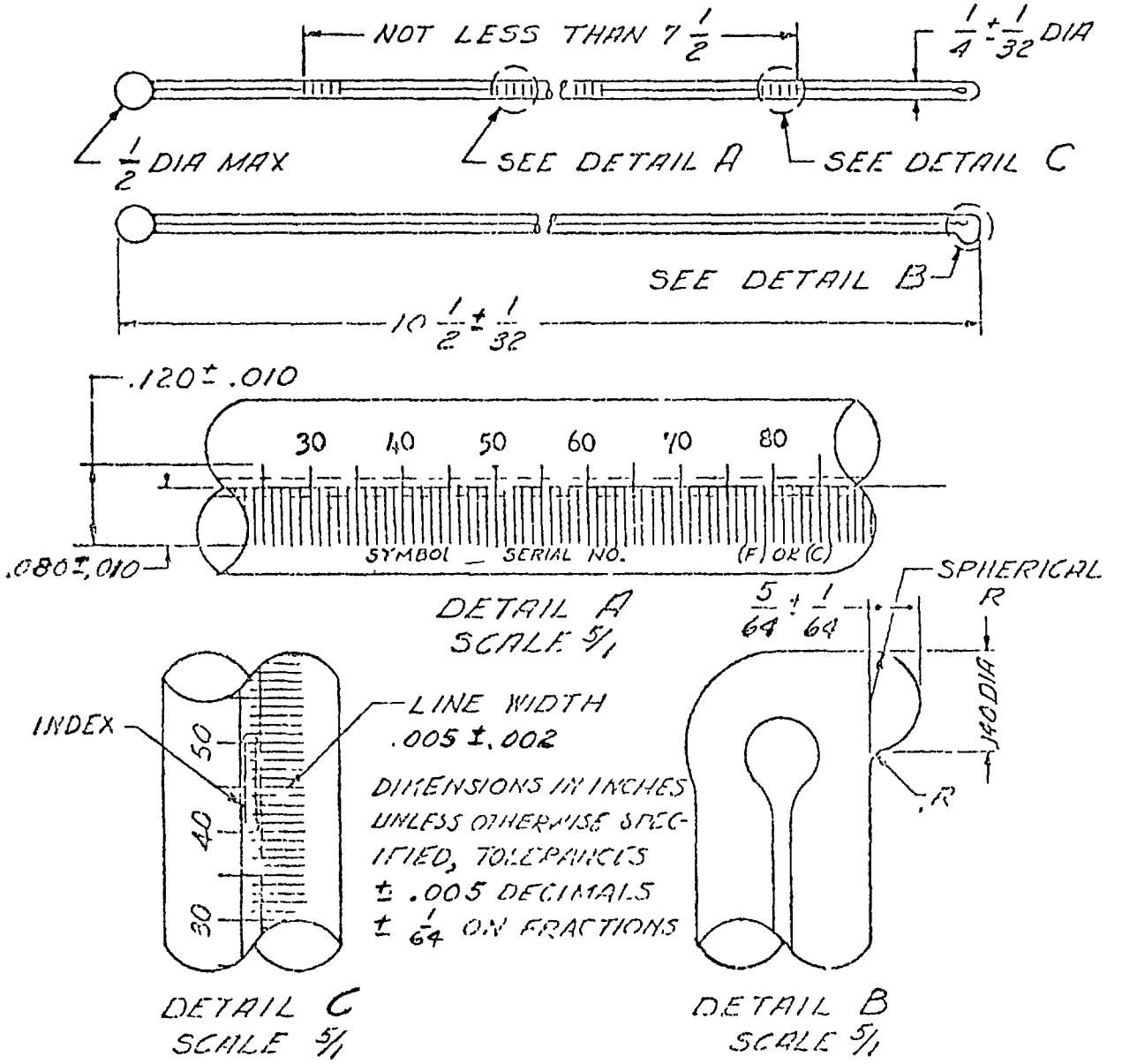
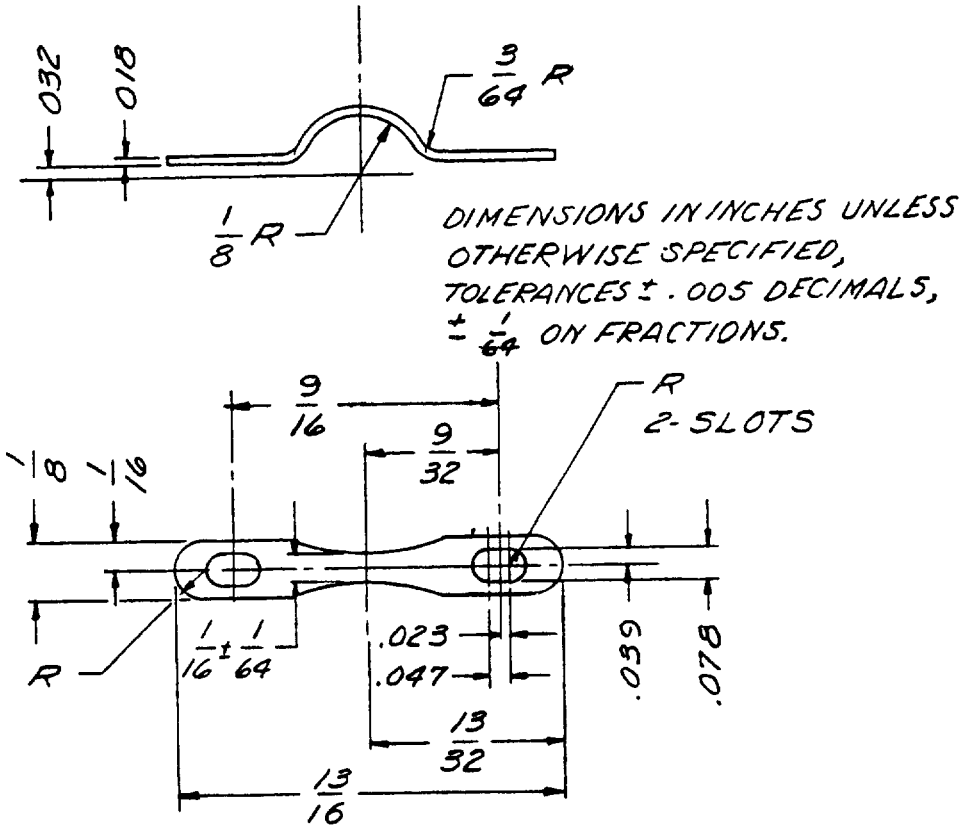


FIGURE 2. Tube.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:50Z
Check the source to verify that this is the current version before use.



Steel, corrosion resisting, class 4, condition C, temper 2, finish 6, per QQ-S-766. (At the option of the ordering office, when so specified in the order (see 6.2), steel shall be corrosion resisting, class 4, condition A, annealed, finish 6, per QQ-S-766.) REMOVE ALL BURRS.

FIGURE 3. Clamp.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:50Z
Check the source to verify that this is the current version before use.

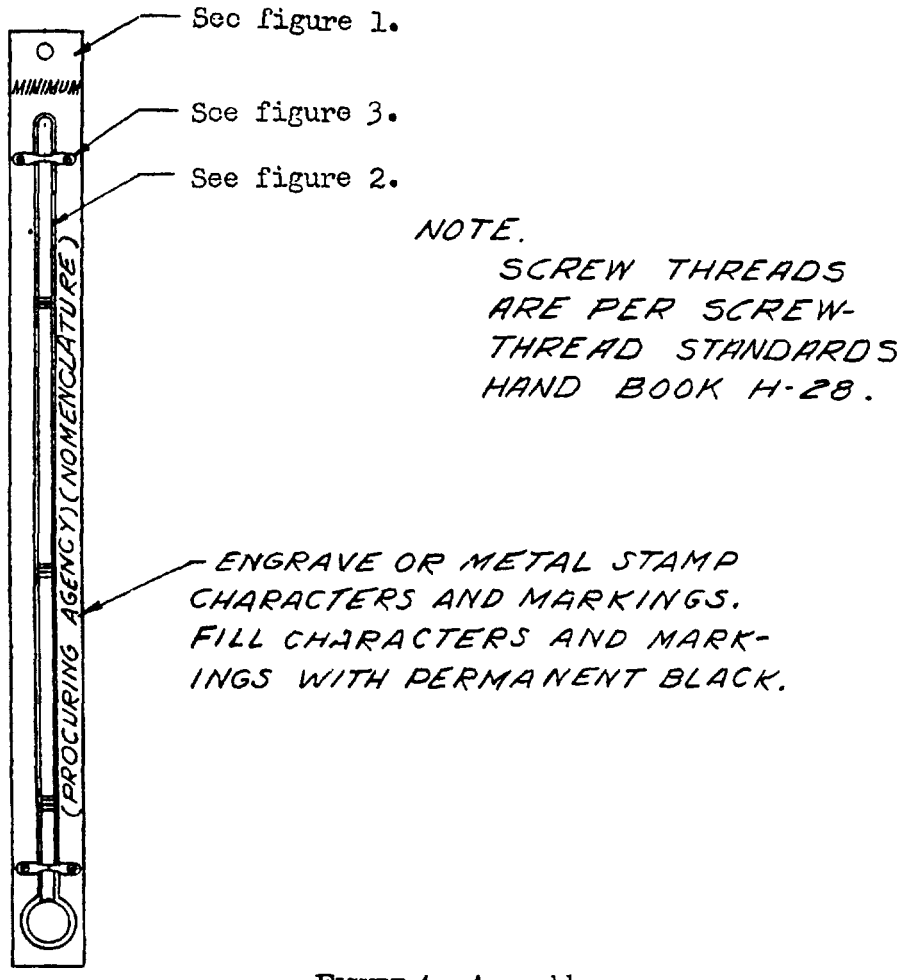


FIGURE 4. Assembly.

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 10 cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:50Z
Check the source to verify that this is the current version before use.